Relief load basis

SO3 Converter Ver 1 · SO3-V1-PE-RLB-001 · Rev 0

Department of Chemical Engineering, Chulalongkorn University (teaching)

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Generated for teaching from the SO3 Converter Ver 1 operator training simulator. Every number comes from the simulator model (design point, noise off) or is marked assumed. Not for construction. No client, contractor or vendor is represented.

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1 · Purpose and basis

This document is the relief load basis of the $SO_2 \rightarrow SO_3$ converter section (K-101 to the stack, with the B-101 steam side and the economizer water side). It follows API 521 for scenario screening and API 520 Part I for sizing (typical practice). Equipment design pressures are taken from SO3-V1-ME-EQL-001; loads come from runs of the simulator model on copies of the design state with the trips switched off.

Method: (1) screen every protected system against the API 521 causes, (2) pick the governing case, (3) size by API 520 and select an API 526 orifice, (4) check inlet loss ≤ 3 % and built-up back pressure ≤ 10 % of set, (5) route the discharge to a safe place.

Gas train (DP-BASIS): the only pressure source is K-101, whose shut-off is 75.0 kPa g at 90 °C air (model) and 81.8 kPa g at 60 °C inlet (assumed). Every part of the gas train is rated 100 kPa g (EQL), so no blocked outlet can over-pressure it and no PSV is fitted to R-201 or the ducts.

2 · Relief device list

Register	Tag	Protected	Set	Scenarios	Governing	Size
PSV-101	PSV-101A / B	B-101 steam drum (steam side of the waste heat boiler)	4.40 MPa g (assumed, = drum design 4.40)	Blocked steam outlet; fire; TV-201 fails closed (more steam); BFW loss; power failure	Blocked steam outlet at the 45 t/h sulfur rate: 103.8 t/h total, 38.9 t/h per drum valve	L (4 × 6 in), 1539 mm² required, 46.6 t/h rated
PSV-101	PSV-101C	SH-201 superheater outlet (lifts first, keeps steam flowing through SH-201)	4.25 MPa g (assumed)	Blocked steam outlet (share 25 %); TV-231 / steam-flow loss overheating SH-201 tubes	Blocked steam outlet: 25.9 t/h at 292 °C	K (3 × 4 in), 1092 mm² required, 28.2 t/h rated
TSV-ECO	TSV-203 / 204 / 205	BFW side of economizers E-203, E-204, E-205	5.50 MPa g (assumed, = economizer design)	Blocked-in BFW heated by gas: liquid expansion, then boiling; BFW loss; tube rupture into the gas	Liquid expansion 110 L/min (E-203) → D orifice; steaming 8.7 t/h is NOT covered (query)	D (¾ × 1 in typical, D area 71 mm²), 327 L/min liquid
TSV-ACID	TSV-301 / 302	Blocked-in acid: IPA and FAT pump discharge between valves, acid cooler acid side	0.70 MPa g (assumed, line design)	Thermal expansion (sun, hot side of the cooler)	Thermal expansion 0.075 L/min (solar 0.6 kW/m² on 10 m², assumed)	¾ × 1 in (D, 71 mm²), 81 L/min
EXD-101	EXD-101	Sulfur furnace F-101 shell	burst 85 kPa g static (assumed)	Deflagration at H-101 light-off; sulfur pool flash after a cold admission	Delayed-ignition puff at start-up (HAZID HZ-01, HAZOP R14/R15)	Vent area by NFPA 68 (not API 520): to be sized
DP-BASIS	— (no device)	K-101 discharge ducts, F-101, B-101 gas side, R-201, E-201…E-205 gas side, T-301/T-302 gas side	design 100 kPa g / −5 kPa g (assumed)	Blocked outlet against the K-101 shut-off; air valves wide open; B-101 tube rupture	K-101 shut-off 75.0 kPa g at 90 °C air; 81.8 kPa g at 60 °C (assumed coldest)	none: rated for the shut-off

3 · Scenario screening (API 521)

G governing, Y applies but smaller, N not credible.

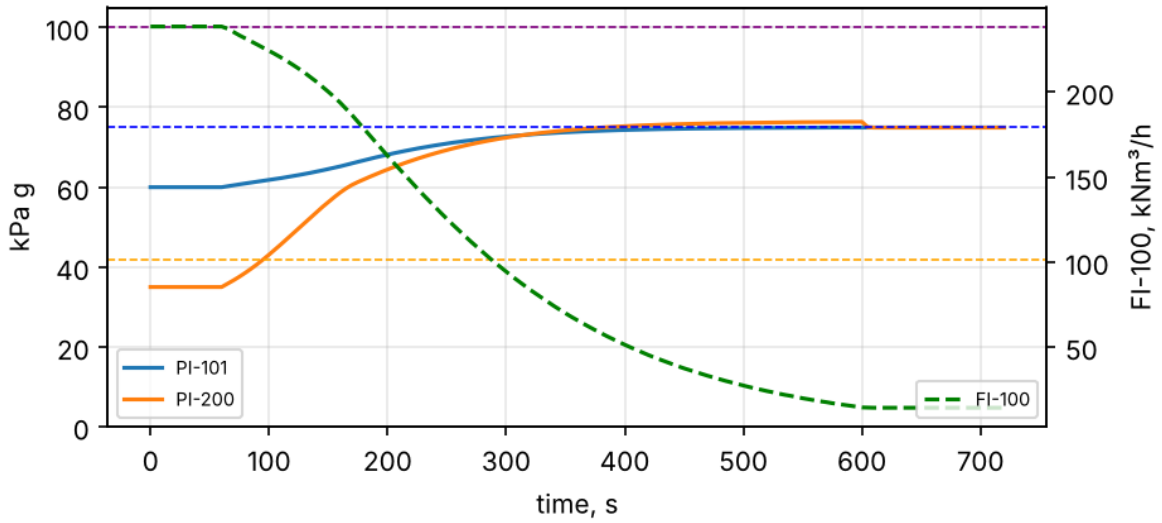
System	Blocked outlet	Fire	Control-valve failure	Cooling / BFW loss	Tube rupture	Power failure	Thermal expansion
B-101 steam drum + SH-201 (PSV-101)	G · Steam header or SH-201 outlet valve closed while hot gas still flows: 103.8 t/h (model, 45 t/h sulfur)	N · Drum elevated 12 m (assumed): no wetted area within 7.6 m of grade; at grade it would be only 1.2 t/h	Y · TV-201 failing closed sends all gas through B-101: 118.0 t/h (model); below the installed capacity	Y · BFW loss: level falls, steam make falls; protects by low-level trip (outside the model), not by relief	N · Fire-tube boiler: a tube failure lets steam into the gas side, not the reverse (see gas train)	N · K-101 stops: gas and heat stop; refractory heat makes little steam	N · Steam drum is not liquid-full in service
Economizers E-203/204/205, BFW side (TSV-ECO)	Y · BFW outlet isolated: same as blocked-in below	N · Inside the gas ducting; no fire exposure	Y · BFW control valve (drum level) closing reduces the flow: see BFW loss	G · BFW at 10 % (model fault, trips off): water outlet 462 °C against Tsat 265 °C, 8.7 t/h of steam in E-203	Y · Water into SO3 gas: acid in the IPA/FAT feed; a corrosion and process case, small pressure effect	Y · BFW pumps stop with K-101 still running: same as BFW loss	Y · Blocked in while hot: 110 L/min expansion (E-203, API 521)
Gas train: ducts, F-101, B-101 gas side, R-201, exchangers, towers (DP-BASIS)	G · Outlet throttled: PI-200 settles at 74.9 kPa g, the K-101 shut-off (model run 1)	N · No flammable inventory inside; external fire on large ducts is not a relief case	Y · FV-101 and FV-102 wide open: PI-200 43.6 kPa g at 266 kNm³/h (model run 2)	N · No cooling medium on the gas side	Y · B-101 tube rupture: 38 kg/s flashing water (order of magnitude) ≈ 18 kPa g on its own, below the shut-off	N · K-101 stops: pressure falls; check vacuum on cool-down instead	N · Gas, not liquid

System	Blocked outlet	Fire	Control-valve failure	Cooling / BFW loss	Tube rupture	Power failure	Thermal expansion
Acid circuits: T-301/T-302 pumps and coolers (TSV-ACID)	Y · Pump against a closed valve: shut-off head below the line rating (assumed)	N · Acid is not flammable	N · Flow valves cannot raise the pressure above pump shut-off	Y · Cooling-water loss makes the acid hot (outside the model), not a pressure case	Y · Cooler leak: acid and water mix; the cooling-water side needs its own thermal relief	N · Pumps stop: pressure falls	G · Blocked-in acid heated by the sun: 0.075 L/min
Molten sulfur line P-101A/B → XV-103 → FV-103 (no relief item)	Y · Pump dead-head: shut-off below the line rating (assumed, HAZOP R06)	Y · Sulfur fire at the pit or pump: outside the battery limit	N · FV-103 failure cannot exceed the pump shut-off	N · —	N · —	N · Pumps stop	G · Steam-jacketed sulfur trapped between XV-103 and FV-103 or pump valves: thermal relief not in the register (query)

4 · Model relief runs

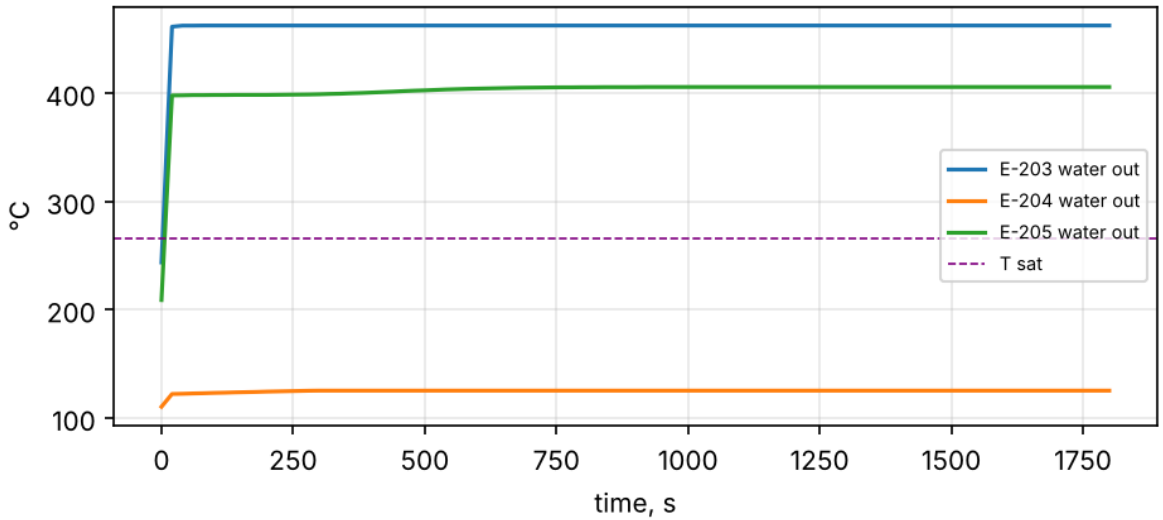
Copies of the design state, dt 0.5 s, loops in AUTO unless stated, trips disabled (no-trip flag and K.trip* lifted in try/finally). Script tabs/erel/tools/gen.js.

4a · Converter outlet throttled until blocked



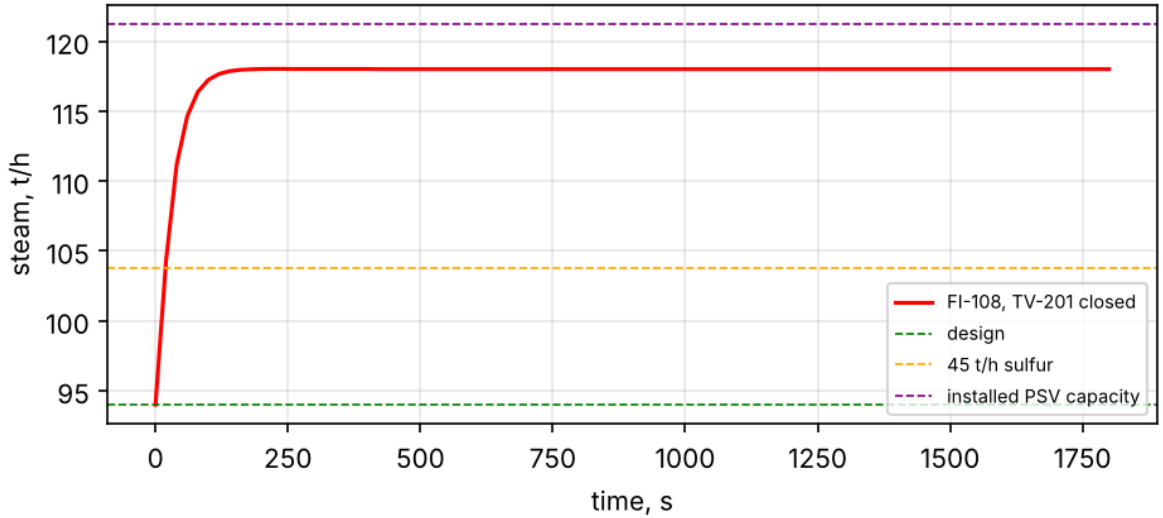
Highest PI-101 75.0 kPa g (= shut-off); PI-200 settles at 74.9 kPa g (peak 76.4 during the ramp, a 2 s flow-lag artefact of the model). With both air valves wide open PI-200 reaches 43.6 kPa g at 266 kNm³/h. FI-100 crosses the FSLL-100 limit at t = 310 s; PI-200 HI alarms at t = 100 s.

4b · Loss of BFW to the economizers



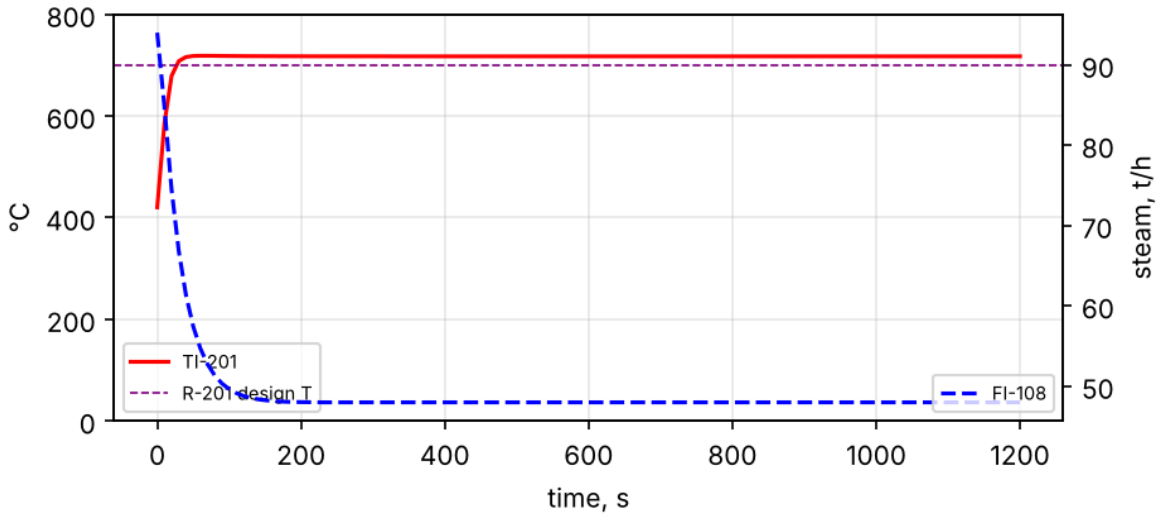
Economizer	Design duty, MW	Duty at 10 % BFW, MW	Water out, °C	Steam made, t/h	Fully blocked in, t/h
E-203	27.7	7.15	462	8.7	15.8
E-204	1.0	0.40	125	0.0	0.9
E-205	20.8	6.01	406	6.2	13.2

4c · Steam make: TV-201 closed and the 45 t/h rate



Case	Steam, t/h	Use
Design FI-108	94.0	normal
45 t/h sulfur	103.8	governing blocked-outlet load
TV-201 failed closed	118.0	capacity check
TV-201 failed open	48.0	not a relief case
Fire on drum	0.0	drum 12 m up (assumed); at grade 1.2

4d · TV-201 fails open (temperature case)



TI-201 reaches 718 °C against an R-201 design temperature of 700 °C; pressure does not change; steam falls to 48.0 t/h. No relief device helps: this is a design-temperature and SIS case.

5 · Sizing (API 520 Part I)

Gas/vapour, critical flow: A = W / (C Kd P1 Kb Kc) · √(TZ/M), C = 0.03948 √(k (2/(k+1))^((k+1)/(k−1))). Steam (Napier): A = 190.5 W / (P1 Kd Kb Kc KN KSH). Liquid: A = 11.78 Q / (Kd Kw Kc Kv) · √(G/(p1 − p2)). Thermal expansion (API 521): q = β Φ / (d c). SI units: A mm², W kg/h, P1 kPa a, T K, Q L/min.

Worked calculation · PSV-101A

- Governing: blocked steam outlet at the 45 t/h rate, 103.8 t/h; SH-201 valve 25 % = 25940 kg/h; two drum valves 38910 kg/h each.
- P1 = 4400 × 1.10 + 101.3 = 4941 kPa a; saturated T 263.1 °C, KSH = KN = 1.
- A = 190.5 × 38910 / (4941 × 0.975) = 1539 mm² (gas equation 1494 mm²) → orifice L (1841 mm², 4 × 6 in), rated 46.6 t/h.
- PSV-101C: 1092 mm² at 4776 kPa a, KSH 0.972 → K, 28.2 t/h. Installed 2 × 46.6 + 28.2 = 121.3 t/h.

Thermal relief

E-203 blocked in with hot gas: liquid expansion 110 L/min against 327 L/min for a D orifice (¾ × 1 in) — liquid covered; steaming 8.7 t/h against 2.23 t/h — not covered (EQ-er-03). Acid between valves: 0.075 L/min against 81 L/min.

6 · Inlet loss, back pressure, disposal

PSV-101A at rated 46.6 t/h: inlet ID 102.3 mm, 1.5 m, ΣK 0.8 → 62.9 m/s, ΔP 50.4 kPa = 1.15 % of set (≤ 3 %). Outlet ID 202.7 mm, 15 m, ΣK 2.5, silencer 15 kPa, isothermal flow, exit choked at 183 kPa a → P2 466 kPa a, built-up back pressure 365 kPa = 8.3 % of set (≤ 10 %, conventional).

Device	Discharges to	Notes
PSV-101A/B/C	Atmosphere via silenced vent stacks	Outlet ≥ 3 m above platforms within ≈ 15 m (typical), weep hole, reaction-force supports
TSV-203/204/205	BFW drain / blowdown tank	Flashing hot water to a drained safe point
TSV-301/302	Closed acid drain or acid bund	98.5 % H_2SO_4: no open discharge
EXD-101	Clear fenced zone beside F-101	Hot gas / SO_2 puff
DP-BASIS	None	Stack is the only outlet of the gas train

7 · Checks and engineering queries

22 ✓ · 4 ✗ of 26 checks (the tab recomputes them).

#	Check		Values
1	Every PSV/TSV item of the Safety-layers register has a governing scenario and a size	✓	4 devices (PSV-101A/B, PSV-101C, TSV-ECO, TSV-ACID)
2	PSV-101A/B set ≤ B-101 drum design pressure	✓	4.40 ≤ 4.40 MPa g (EQL)
3	PSV-101C set ≤ SH-201 design pressure	✓	4.25 ≤ 4.40 MPa g (EQL)
4	PSV-101C (SH-201) lifts before the drum valves	✓	4.25 + 0.10 (SH ΔP) < 4.40 MPa g
5	Drum operating pressure ≤ 90 % of the lowest drum PSV set (simmer margin)	✗	4.00 / 4.40 = 90.9 %
6	Installed steam capacity ≥ governing load (blocked outlet, 45 t/h sulfur)	✓	121.3 ≥ 103.8 t/h
7	Installed steam capacity ≥ TV-201 failed-closed steam make	✓	121.3 ≥ 118.0 t/h (model run 4)
8	SH-201 valve carries ≥ 20 % of the total (typical boiler practice)	✓	23.2 %
9	TSV-ECO set ≤ economizer design and above its operating pressure	✓	5.00 < 5.50 ≤ 5.50 MPa g (EQL)
10	TSV-ECO (D) passes the liquid expansion of the largest economizer	✓	327 ≥ 110 L/min
11	TSV-ECO (D) passes the steaming load of a blocked-in economizer	✗	2.23 t/h vs E-203 8.7, E-204 0.0, E-205 6.2 t/h
12	TSV-ACID passes the thermal expansion load	✓	81 ≥ 0.075 L/min
13	K-101 shut-off (model, 90 °C air) ≤ gas-train design pressure	✓	75.0 ≤ 100 kPa g (EQL)
14	K-101 shut-off at the coldest blower inlet ≤ gas-train design pressure	✓	81.8 kPa g at 60 °C ≤ 100
15	Highest PI-200 in the model runs ≤ R-201 design pressure	✓	76.4 ≤ 100 kPa g (peak during the ramp; settles at 74.9)
16	PI-200 HI alarm sits below the gas-train design pressure	✓	alarm 42 < 100 kPa g
17	PI-200 HI alarm sits above PI-200 at the highest normal rate (45 t/h sulfur)	✗	alarm 42 vs 42.3 kPa g at 45 t/h (model)
18	B-101 tube rupture: steam alone stays below the K-101 shut-off	✓	18.0 < 75.0 kPa g (order of magnitude)
19	EXD-101 bursts above the K-101 shut-off (no SO_2 release on a blocked outlet)	✓	85 > 81.8 kPa g
20	EXD-101 bursts below the F-101 design pressure	✓	85 < 100 kPa g
21	R-201 / bed-1 inlet design temperature covers TV-201 failing open	✗	TI-201 up to 718 °C vs design 700 °C (EQL R-201)
22	P-101A/B shut-off ≤ sulfur line design pressure (no relief needed for dead-head)	✓	720 ≤ 1100 kPa g (EQL)
23	P-301/P-302 shut-off ≤ acid circuit design pressure	✓	456 ≤ 700 kPa g (EQL)
24	EQL K-101 shut-off agrees with the model run	✓	EQL 75 vs model 75.0 kPa g
25	PSV-101A inlet loss ≤ 3 % of set	✓	1.15 %
26	PSV-101A built-up back pressure ≤ 10 % of set (conventional valve)	✓	8.3 %

Engineering queries

Id	Subject	Finding	Proposed	Disc.
EQ-er-01	R-201 and the gas train have no relief device	The converter, F-101, B-101 gas side, the exchangers and the towers rely on DP-BASIS: K-101 shut-off is 75.0 kPa g at 90 °C (model) and 81.8 kPa g at 60 °C blower inlet (assumed); the throttled-outlet run settles PI-200 at 74.9 kPa g. The pressure-vessel margins rule (op + 1.7 bar) would give 205 kPa g for R-201, far more than the shut-off needs on a 13 m vessel; the EQL uses 100 kPa g.	Write the gas-train design basis as: design pressure ≥ K-101 shut-off at the coldest blower inlet plus 10 % (≥ 90 kPa g; 100 kPa g assumed) and a vacuum rating for cool-down with the dampers shut (−5 kPa g assumed). Get the certified K-101 curve to confirm the shut-off.	Process
EQ-er-02	Steam PSV set against drum design	With the drum design at 4.40 MPa g (EQL / margins rule on 4.0 MPa g) and the PSVs set at it, the drum works at 90.9 % of the PSV set: above the 90 % usually kept so that spring valves do not simmer.	Raise the drum and SH-201 design pressure to ≈ 4.6 MPa g (operating ≈ 87 % of set) or confirm pilot-operated valves; keep PSV-101C on SH-201 set lowest.	Mechanical
EQ-er-03	Blocked-in economizer can steam	With BFW at 10 % of design (model) the E-203 water outlet reaches 462 °C, above Tsat 265 °C, making ≈ 8.7 t/h of steam; a ¾ × 1 in TSV passes 2.23 t/h of steam.	No isolation valve between the economizers and the drum, or car-sealed open; the TSVs cover only liquid expansion during maintenance. Low BFW-flow alarm FI-253 is already in ALM; add the economizer outlet temperature to the steam-system P&ID.	Process
EQ-er-04	TV-201 failing open: temperature basis of the bed-1 inlet duct	TV-201 wide open drives TI-201 to 718 °C (model, trips off) against an R-201 / bed-1 inlet design temperature of 700 °C.	Design the B-101 outlet/bed-1 inlet duct and the R-201 top shell (or their refractory) for the TV-201 open case, or limit TV-201 travel with a mechanical stop sized from the model.	Mechanical
EQ-er-05	EXD-101 burst band	An explosion panel on F-101 must burst above the K-101 shut-off (81.8 kPa g at 60 °C) or a blocked outlet releases SO_2, yet below the shell strength; 85 kPa g (assumed) leaves a narrow band under 100 kPa g.	Decide: rate F-101 for the start-up deflagration and drop EXD-101, or size EXD-101 by NFPA 68 with Pstat in the band and a safe discharge area; keep the H-101 purge (HAZOP R15) either way.	Process
EQ-er-06	Molten sulfur trapped between valves	Steam-jacketed sulfur between XV-103 and FV-103 (and between pump valves) can be blocked in and heated; the Safety-layers register has no thermal relief for it.	Add thermal relief (jacketed, to the sulfur pit) or a procedure that never blocks in a heated section; confirm in the HAZOP close-out.	Process
EQ-er-07	PI-200 HI alarm at the top of the normal range	At 45 t/h sulfur (model, loops in AUTO) PI-200 is 42.3 kPa g, above the HI alarm of 42 kPa g; with both air valves wide open it reaches 43.6 kPa g. The alarm would stand during a rate push.	Review the PI-200 HI alarm (e.g. 45 kPa g, still far below the 100 kPa g design) in the alarm rationalisation, or limit the maximum rate.	Instrument
EQ-er-08	Steam side is outside the model	Steam loads come from the model gas side (FI-108 94.0 t/h design, 103.8 t/h at 45 t/h sulfur, 118.0 t/h with TV-201 closed); drum size, elevation and SH-201 pressure drop are assumed.	Issue the steam-system P&ID and drum datasheet; re-run this basis with the real drum and SH-201 data.	Process

8 · Assumptions

Item	Value	Source
B-101 drum operating / design	4 / 4.4 MPa g	EQL
Drum PSV sets A/B, SH PSV set	4.4 / 4.4 / 4.25 MPa g	assumed
SH-201 ΔP, steam relief share	0.1 MPa, 25 %	assumed
Accumulation; Kd gas, liquid	10 %; 0.975, 0.65	API 520 typical
Drum size, elevation	1.6 m × 8 m, 12 m	assumed
Economizer op / design	5 / 5.5 MPa g	EQL
Acid design	0.7 MPa g	EQL
Gas-train design / vacuum	100 / -5 kPa g	EQL
Coldest blower inlet	60 °C	assumed
R-201 design temperature	700 °C	EQL
EXD-101 static burst	85 kPa g	assumed
Tube ID, flashing flux	45 mm, 12000 kg/(m² s)	assumed
Water / acid β, ρ, c	0.001, 865, 4500 / 0.00056, 1800, 1500	assumed
Solar flux, exposed acid area	0.6 kW/m², 10 m²	assumed